GRAND ROUNDS CASE · FOLLOW-UP PROGRAM

Followup 26-13

Wide-complex tachycardia and fusion beats in a device patient

Patient	61-year-old male. Heart failure with reduced ejection fraction, most recently 20 to 25 percent, down from 55 percent a year earlier. Implanted cardioverter-defibrillator placed in 2023. COPD on 2 L home oxygen, type 2 diabetes, chronic kidney disease stage 3b with a baseline creatinine around 2.0, obstructive sleep apnea on CPAP, hypertension, and class 3 obesity with a body mass index of 74. **Had run out of his loop diuretic** and had been weak and increasingly short of breath for several weeks.
Outcome	Survived. Acute on chronic heart failure, acute and chronic hypercapnic respiratory failure, and acute kidney injury. CT angiography excluded pulmonary embolism and aortic dissection and showed pleural effusions with central vascular congestion. Troponin flat. Pro-BNP 10,098, double his previous value.
Disposition	BiPAP in the emergency department, then aggressive diuresis over several days with high-dose loop diuretic, metolazone, and acetazolamide. Discharged home with services.

The call

Dispatched priority 2 for weakness.

He was supine in bed, pink, warm and dry, alert and oriented times four, with **audible labored breathing**. He described weakness for several weeks and could not say exactly how long. He had run out of his diuretic and was short of breath. **Pitting edema bilaterally from the feet to the knees.** Wheezing in both upper and lower lobes. He sat up with the help of his walker and walked to the front door.

Course, relative to patient contact:

From contact	Finding or action
+0	Airway, breathing, circulation. Neurologic assessment. Full assessment.
+12	**Blood pressure unobtainable** by the monitor. Pulse 120, respirations 20, SpO2 98%. Glucose 109.
+12	Cardiac monitor. Rate varying between 90 and 120.
+13	**12 lead ECG.**
+17	Left scene. Manual blood pressure attempted, also unobtainable.
+20	SpO2 98%.
+23	**12 lead ECG.**
+24	Arrived at the receiving facility. SpO2 98%.

En route, the device began producing pacing activity, which the crew described as firing approximately six times, and the rhythm then became consistently paced. **The transport was upgraded to priority 1.** Intravenous access was attempted and not obtained. At arrival he was unable to sign for himself, and the receiving nurse signed.

First tracing, +13 minutes. Rate 124. QRS 0.154 seconds. PR 0.196. Axis -11 degrees. The monitor's interpretation: **probable ventricular tachycardia**, intraventricular conduction defect, possible anterior infarct of undetermined age, inferior and lateral ST-T abnormality possibly due to ischemia, low QRS voltages in the precordial leads. The crew's narrative reads the rhythm as sinus tachycardia.

Second tracing, +23 minutes. Rate 115. QRS 0.132 seconds. The monitor's interpretation: **demand pacing**, pacemaker rhythm, no further analysis. Pacing spikes visible across the rhythm strip.

What the crew saw between those two tracings, confirmed in conversation after the call, was **complexes changing shape from beat to beat with pacing spikes appearing in front of some and not others**, followed by the rhythm settling into consistent pacing. Those are fusion beats.

Primary impression

Decompensated heart failure after running out of diuretic, with a wide-complex tachycardia and device activity en route.

The heart failure part is straightforward and the hospital confirmed it. He had stopped his loop diuretic, fluid accumulated over weeks, and it filled his legs, his abdomen, his pleural spaces, and the interstitium of his lungs. The pro-BNP was 10,098 against a ceiling of 125.

The rhythm part is not settled, and the record cannot settle it. Three readers looked at the first tracing. The monitor called probable ventricular tachycardia. The crew called sinus tachycardia. The emergency physician, looking at the department's own ECG, documented sinus tachycardia without ST abnormality. The features on the tracing cut in both directions, and the education section works through them.

The device was never interrogated. No note in the admission records a device check, and the pacing activity the crew saw does not appear in any hospital note. Interrogation is the one test that would have answered what the rhythm was and how the device classified it, and it was not done.

The upgrade to priority 1 was the right call. A crew watching a device patient's rhythm change shape in front of them, without a name for what they are seeing, upgraded on an unidentified change in a cardiac patient with an ejection fraction of 20 percent. That is exactly the correct instinct.

Hospital course

Emergency department. Triage found him at 124 with respirations of 21, saturating **91 percent** and needing 28 percent oxygen, blood pressure recorded at 155/127. He was in respiratory distress with coarse breath sounds, **3+ pitting edema** in both legs with chronic venous stasis changes, and a distended abdomen. He was placed on **BiPAP**.

The working differential was acute heart failure, COPD, and pulmonary embolism. A venous blood gas showed a pH of 7.30 with a PCO2 of 71, hypercapnic respiratory failure. The chest radiograph was concerning enough that CT angiography was ordered for dissection, and it found **no dissection and no pulmonary embolism, but pleural effusions and central vascular congestion.** Pro-BNP 10,098, double his previous value. Creatinine 2.54, up from a baseline around 1.9. Sublingual nitroglycerin was given for persistently elevated pressures and intravenous diuresis started.

By early evening his tachycardia had come down, he was responding to BiPAP and diuretic with good urine output, and he was admitted.

Hospital days 1 through discharge. High-dose intravenous loop diuretic three times daily, with metolazone added to overcome diuretic resistance. When his bicarbonate climbed from the diuresis, metolazone was held and acetazolamide given to correct it. Sacubitril-valsartan was held for the rise in creatinine; spironolactone and dapagliflozin were continued. CPAP overnight. Iron deficiency identified, with an iron saturation of 9 percent, and intravenous iron ordered. Physical therapy found him walking 20 feet with a walker, limited by breathlessness, and recommended discharge home with services.

Final diagnoses: acute on chronic heart failure with reduced ejection fraction; acute and chronic hypercapnic respiratory failure; acute kidney injury on chronic kidney disease stage 3b; iron deficiency anemia; obstructive sleep apnea; COPD.

Education points

Ventricular tachycardia or aberrancy

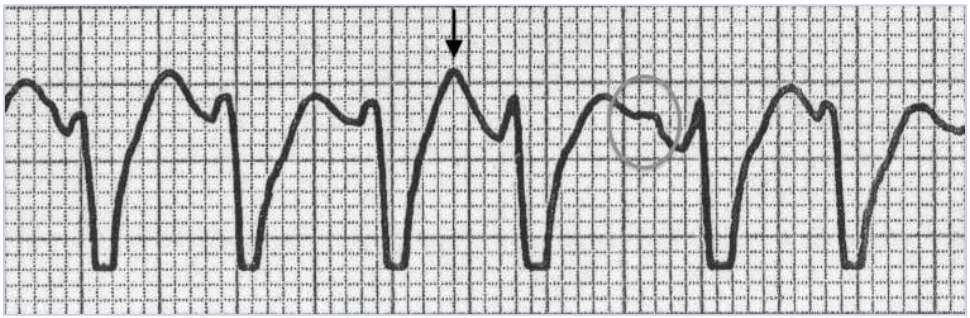
A wide-complex tachycardia is any rhythm faster than 100 with a QRS longer than 0.12 seconds. Most are one of two things. **Ventricular tachycardia**, where the impulse originates in the ventricle and spreads through muscle rather than the conduction system, which is slow and produces a wide complex. Or **supraventricular tachycardia with aberrant conduction**, where the impulse starts above the ventricles normally but meets a blocked or diseased bundle branch on the way down, so it spreads the slow way and looks wide for a different reason.

The distinction matters because the treatments diverge and the wrong one can kill. A calcium channel blocker given to a patient in ventricular tachycardia removes the contractility and vascular tone that are holding a pressure up, and can cause collapse.

The single most useful predictor is not on the ECG. It is the patient. Roughly 80 percent of wide-complex tachycardias are ventricular, and in patients with structural heart disease or a prior infarct the proportion is higher still. This patient had an ejection fraction of 20 to 25 percent and a possible old anterior infarct on the tracing. That alone moves the probability substantially toward VT before anybody looks at a single lead.

ECG features that favor ventricular tachycardia:

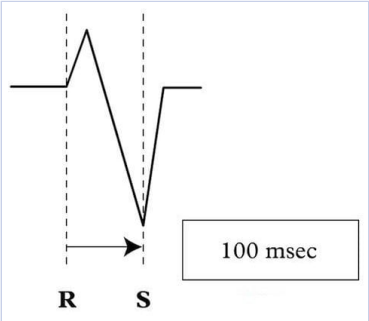
- **AV dissociation.** P waves marching through at their own rate, unrelated to the QRS complexes. This is close to diagnostic when you can see it.



AV dissociation in ventricular tachycardia. The circled notch is a P wave landing at its own rate, unrelated to the wide complexes. Image: Life in the Fast Lane, litfl.com, CC BY-NC-SA 4.0.

- **Capture and fusion beats**, which suggest AV dissociation and are covered in the next section. They appear in only about one VT in five, so their absence tells you little, and on their own they are not diagnostic.

- **Very wide complexes.** A QRS above 0.14 seconds with a right bundle pattern, or above 0.16 with a left bundle pattern.
- **Extreme axis**, in the northwest quadrant between -90 and 180 degrees. No supraventricular rhythm conducts in that direction.
- **Precordial concordance**, meaning every complex from V1 to V6 points the same way, all up or all down.
- **An initial R wave in lead aVR**, from the Vereckei approach.
- **Brugada's criteria**, a stepwise algorithm: absence of any RS complex across the precordial leads, or an interval from the start of the R wave to the bottom of the S wave longer than 100 milliseconds in any precordial lead, favors VT.



Measuring the RS interval: from the start of the R wave to the lowest point of the S wave. Longer than 100 milliseconds in any precordial lead favors VT. Image: Life in the Fast Lane, litfl.com, CC BY-NC-SA 4.0.

Features that favor aberrancy:

- **A prior ECG showing the same wide morphology in sinus rhythm.** This is the most powerful evidence for aberrancy available. If the wide complex during the tachycardia is identical to the patient's known baseline bundle branch block, SVT with aberrancy can be called with real confidence.
- A normal or near-normal axis.
- A recognisable bundle branch block morphology.

Two things that do not distinguish them. Hemodynamic stability, because stable VT is common and plenty of patients walk in with it. And response to adenosine, which is unreliable.

On the first tracing, the features cut both ways. The axis of -11 degrees favors aberrancy. The machine measured a consistent PR interval of 0.196, which if accurate argues against AV dissociation, though machine PR measurement in a wide rhythm is unreliable. He has a documented intraventricular conduction defect, which means his baseline QRS is wide and aberrancy is expected. Against that: an ejection fraction of 20 to 25 percent, a possible prior infarct, a QRS of 0.154, and a monitor that called VT.

It cannot be settled from a compressed image. The piece of evidence that would settle it, his old ECG in sinus rhythm, was not available in the truck, and the device interrogation that would have told us what the device thought was never done.

Experts disagree on these tracings routinely. Brugada's algorithm performed with near-perfect accuracy in its original study, and when emergency physicians and cardiologists later applied it to real tracings, accuracy dropped substantially and emergency physicians disagreed with each other about one time in five even while using the same rule. Three competent readers disagreeing about this strip is not three people getting it wrong. It is the normal performance of the test.

A wide-complex tachycardia in a patient with structural heart disease is ventricular tachycardia until somebody proves otherwise. You will occasionally treat aberrancy as VT, and that error is survivable. The reverse is not.

> **Further reading on VT and aberrancy.** The ECG criteria are much easier to learn with example tracings beside them. This free reference walks through each one with examples: https://litfl.com/vt-versus-svt-ecg-library/

Fusion beats

A fusion beat is two impulses depolarizing the ventricle at the same moment. The ventricle can be activated from more than one place. If one impulse arrives from above through the normal conduction system and another arrives from somewhere else at nearly the same instant, each depolarizes part of the muscle before they collide in the middle. The resulting QRS complex is a hybrid: part one shape, part the other, and usually narrower than either alone.

There are two very different situations that produce them, and they mean opposite things.

Pacing fusion. A pacemaker delivers its impulse at almost exactly the moment the patient's own impulse arrives from above. Part of the ventricle is activated by the device, part by the native beat. On the monitor you see a **pacing spike followed by a complex that looks different from both a fully paced beat and a fully native beat.**



Normal sequential pacing. A small atrial pacing spike, then a ventricular pacing spike, before every wide complex. Every beat is captured. Image: Life in the Fast Lane, litfl.com, CC BY-NC-SA 4.0.



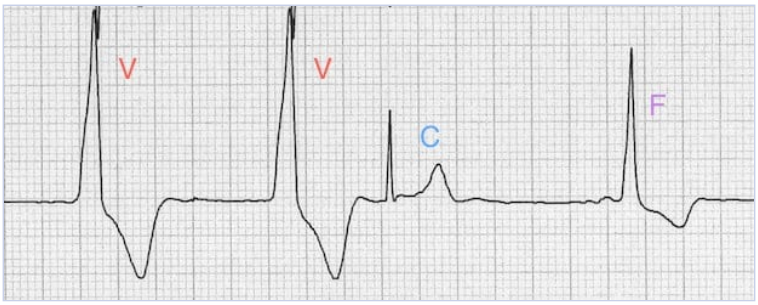
Pacing fusion. P marks fully paced beats. F marks fusion beats, where the native impulse collides with the paced one and the complex comes out smaller and narrower. The beat marked ?C is probably a native capture beat. Image: Life in the Fast Lane, litfl.com, CC BY-NC-SA 4.0.

This happens when the device's pacing rate and the patient's own rate are close together and competing. As one or the other wins, the complexes change shape beat by beat, with some fully native, some fully paced, and some hybrid in between. When the native rate falls below the device's threshold, the device takes over completely and you see consistent paced rhythm, which is what the second tracing shows.

Pacing fusion is usually reassuring. It means the device is sensing the native rhythm, timing its output against it, and capturing. It is a device working as designed, and the transition the crew saw is what a pacemaker competing with a native rhythm at a similar rate looks like.

Native fusion is the opposite. In a patient in ventricular tachycardia, a sinus impulse occasionally gets down through the AV node and partially activates the ventricle at the same time the ventricular focus fires. That produces a hybrid complex **with no pacing spike in front of it**, in the middle of a wide-complex tachycardia.

Native fusion suggests the atria and ventricles are running independently, which is AV dissociation, and **AV dissociation is highly specific for ventricular tachycardia.** The fusion beat is evidence of that dissociation rather than proof of VT: fusion and capture beats are not diagnostic on their own and can occur in supraventricular rhythms with aberrant conduction.



Native fusion, with no pacing spikes anywhere. V marks wide ventricular complexes, C a narrow capture beat conducted from the atria, and F a fusion beat between the two. This strip is an accelerated idioventricular rhythm rather than VT, but the fusion mechanism is identical. Image: Life in the Fast Lane, litfl.com, CC BY-NC-SA 4.0.

The difference between the two is the spike. A pacing spike before the hybrid complex means pacing fusion. No spike means native fusion, which in a wide-complex tachycardia raises the probability of VT.

Pseudofusion is the close relative. The device fires, but the native impulse has already depolarized the ventricle, so the spike lands on a QRS that is entirely native. The complex looks normal for the patient with a spike sitting on it. It is harmless and it means the device fired a fraction too late to contribute.

When a device patient's rhythm does something that should worry you:

- **Failure to capture.** A pacing spike with no complex after it. The device fired and the heart did not respond.
- **Failure to sense.** The device paces on top of the patient's own beats as if it cannot see them.
- **Pacing at an unexpected rate**, or pacing that does not match the underlying rhythm.
- **A defibrillator discharge**, which the patient will feel and describe unmistakably.

In this case the crew saw fusion transitioning to consistent pacing, and the second tracing confirms the device capturing. That is the reassuring version. What the record cannot show is which kind of fusion the transitional beats were, because they happened between the two tracings and nobody knew to look for the spike.

Why a defibrillator may not treat ventricular tachycardia

This patient has an implanted cardioverter-defibrillator, not only a pacemaker. An implanted defibrillator is also a pacemaker, but not every pacemaker is a defibrillator. The difference matters on scene because a defibrillator can treat ventricular tachycardia itself.

It does that in two ways. First, with **antitachycardia pacing**: rapid bursts of pacing faster than the tachycardia, which can interrupt a reentrant circuit painlessly. The patient often feels nothing, or a flutter. If that fails, it charges and delivers a **shock**, which the patient feels as a violent blow to the chest.

Every defibrillator is programmed with a detection zone. It only treats rhythms faster than a cutoff rate set for that individual patient, commonly somewhere well above 150. Below the cutoff it watches, and does nothing.

So a slow ventricular tachycardia can run untreated in a patient with a working defibrillator. A rate of 124 is slow for VT, and slow VT is common in cardiomyopathy, especially in patients on drugs that slow conduction. If

this patient's rhythm on the first tracing was VT at 124, a device with a typical detection cutoff would have classified it and left it alone.

Interrogation answers it. The device stores what it detected, how it classified each episode, and electrograms of the rhythm from inside the heart. A device check answers what a surface ECG cannot. It was not done on this admission.

Decompensated heart failure and the cardiac wheeze

He wheezed in every lung field, and it was not his COPD.

Fluid backing up into the lungs does not only flood the alveoli. It first fills the tissue around the small airways and swells their walls, narrowing them. Narrowed airways whistle on expiration exactly the way bronchospasm does. The result is **cardiac asthma**, a wheeze produced by heart failure rather than by airway disease.

In a patient who carries both diagnoses, that is a genuine trap. A wheezing COPD patient gets bronchodilators. A wheezing heart failure patient needs the fluid off and the pressure down, and in a patient with an ejection fraction of 20 percent, the tachycardia and beta-agonist stimulation from repeated nebulizers make the heart work harder for no benefit.

The findings that separate them here were all in the crew's own narrative: he had run out of his diuretic, he had pitting edema from feet to knees, he had been getting progressively weaker and more breathless over weeks. That is fluid, and the hospital's pro-BNP of 10,098 and pleural effusions on CT confirmed it.

Prehospital considerations

Document what the machine interprets, but only use it as a guide. The crew acquired two 12 leads ten minutes apart, before and after the rhythm changed, which is exactly the record that lets anyone analyze this case at all. The first one carries a machine read of probable ventricular tachycardia. **When the machine calls something you disagree with, say both at handoff**: what the monitor read and what you read. The receiving team can decide, but only if they hear it.

Describe the device activity in plain words. "The pacemaker started firing" does not tell a receiving clinician whether it paced, whether it shocked, or what the rhythm did. What they need is what you saw: the complexes changing shape, spikes in front of some and not others, how long it lasted, what the rhythm was before and after, and whether the patient felt anything. The pacing activity on this call does not appear anywhere in the hospital record. A plain description is what gets a device interrogated.

Find out what the device is. A defibrillator and a pacemaker behave differently and mean different things. The device card, the patient, or the family can usually tell you, and the answer changes what you are looking at on the monitor.

Ask for the old ECG. It is the single most powerful piece of evidence for telling VT from aberrancy, and patients with known conduction disease often carry one or have one on a device card or phone. Even a photograph of a prior tracing is useful.

An unobtainable blood pressure in a very large patient is often the cuff. At a body mass index of 74, a standard adult cuff cannot occlude the artery properly. Automated and manual attempts both failed on this call, and the hospital later recorded a pressure of 155/127. A cuff too small for the arm reads falsely high or not at all. A thigh cuff on the upper arm, or a forearm reading, will often work where the standard cuff fails.

The interventions

PREHOSPITAL INTERVENTIONS

Two 12 lead ECGs, ten minutes apart. The most valuable thing done on this call. The first captured a wide-complex tachycardia and the second captured the device-paced rhythm that followed. Between them they bracket the transition, and without them there would be nothing to analyze.

Continuous cardiac monitoring, which is how the crew saw the fusion beats and the transition to pacing. The monitor showed a rate varying between 90 and 120 before the device took over.

Blood pressure, attempted by automated cuff and manually, and unobtainable by both. Covered above. In a patient with a body mass index of 74, cuff size is the likeliest reason.

Glucose, 109. Excluded hypoglycemia as a cause of weakness.

Upgrade to priority 1. On an unidentified change in rhythm in a device patient with a severely reduced ejection fraction. Correct.

Intravenous access, attempted and not obtained. In a patient of this size, peripheral access is genuinely difficult. He needed nothing that required it before arrival.

IN-HOSPITAL INTERVENTIONS

BiPAP. Noninvasive positive pressure did two jobs here. It supported ventilation in a patient with a PCO2 of 71, and it reduced preload and afterload on a failing left ventricle by raising intrathoracic pressure, which is part of why his tachycardia came down.

Sublingual nitroglycerin, for persistently elevated pressures. A venodilator that reduces preload, and at higher doses afterload, taking load off the failing ventricle.

CT angiography of the chest, ordered for possible dissection. It excluded dissection and pulmonary embolism and showed the effusions and vascular congestion of fluid overload.

High-dose loop diuretic with metolazone. Loop diuretics block sodium reabsorption in the loop of Henle. In a patient who has been on them long term, the kidney adapts by reabsorbing more sodium further down the nephron, which blunts the effect. Metolazone blocks that downstream site, so the two together produce far more diuresis than either alone. The strategy is called sequential nephron blockade.

Acetazolamide, when his bicarbonate rose. Aggressive loop diuresis causes a contraction alkalosis, as chloride is lost and bicarbonate is retained. Acetazolamide makes the kidney excrete bicarbonate, correcting the alkalosis and adding some diuresis of its own.

Holding sacubitril-valsartan. A cornerstone heart failure drug, held because it can worsen kidney function when creatinine is already rising. It is a temporary trade of long-term benefit against short-term kidney protection.

Intravenous iron. His iron saturation was 9 percent. Iron deficiency worsens heart failure symptoms and exercise tolerance independent of anemia, and correcting it improves both.

The labs, and what they mean

Reference ranges vary between institutions and analyzers. These are the ranges printed on the source reports.

Heart failure and injury

Lab	Value	Reference range
Pro-BNP (ng/L)	10,098	125 or less
hs troponin T, baseline (ng/L)	54	23 or less
hs troponin T, 2 hour (ng/L)	50	Less than 23
Creatinine (mg/dL)	2.54	0.70 to 1.20
eGFR (mL/min/1.73 m2)	28	60 or greater
BUN (mg/dL)	42	8 to 23

Pro-BNP is released by ventricular muscle when it is stretched. It measures wall tension rather than fluid directly, which is why it is so useful in heart failure. His was 10,098, roughly eighty times the upper limit and double his own previous value. That is a ventricle under enormous volume and pressure load.

The troponin was elevated and flat. 54 at baseline and 50 two hours later, a change of minus 4. An elevated but unchanging troponin is chronic myocardial injury, which is common in advanced heart failure and in chronic kidney disease, both of which he has. It is not the rising pattern of an acute infarct. The possible anterior infarct on the first tracing was read as of undetermined age, and nothing in the troponin suggests it was new.

The creatinine is cardiorenal. 2.54 against a baseline around 1.9. When the heart fails to pump forward and venous pressure backs up, the kidney is squeezed from both sides: less blood arriving and higher pressure in the veins it drains into. The fix is to take the fluid off, which is why the team kept diuresing despite the creatinine, and why sacubitril-valsartan was held rather than the diuretic.

Ventilation and blood count

Lab	Value	Reference range
pH, venous	7.30	7.32 to 7.43
PCO2, venous (mmHg)	71	41 to 51
Total CO2, venous (mmol/L)	31.2	23.0 to 30.0
Hemoglobin (g/dL)	11.8	14.0 to 18.0
Mean cell hemoglobin (pg)	24.3	27.0 to 33.0
Red cell distribution width (%)	16.9	11.0 to 16.0
Potassium (mmol/L)	5.1	3.4 to 5.0

His blood gas shows hypercapnic respiratory failure layered on a chronic problem. A PCO2 of 71 with a pH of 7.30. If a PCO2 that high had developed suddenly, the pH would be far lower. The fact that it is only mildly acidic means his kidneys had already been retaining bicarbonate for a long time to compensate, which is what the total CO2 of 31.2 shows. That is chronic CO2 retention from obesity, sleep apnea and COPD, with an acute worsening on top from the fluid.

That also explains his saturation of 98 percent in the truck against 91 percent at triage and a PCO2 of 71. **A good saturation says nothing about carbon dioxide.** A patient can oxygenate well while retaining CO2 to dangerous levels, and in a patient with chronic retention, capnography would have shown it where the pulse oximeter could not.

The blood count is iron deficiency. A hemoglobin of 11.8, a low mean cell hemoglobin, and a wide red cell distribution width is the pattern of small, pale red cells of varying size, and his iron saturation of 9 percent confirmed it.

The potassium of 5.1 is from a hemolyzed specimen, which pushes potassium up artificially as red cells rupture in the tube. The real value was probably lower.

Learning points

1. **A wide-complex tachycardia in a patient with structural heart disease is VT until proven otherwise.** This patient had an ejection fraction of 20 to 25 percent and a possible old infarct, which makes VT more likely before any lead is examined. The most useful ECG evidence for aberrancy is a prior tracing showing the same wide shape in sinus rhythm, and it is worth asking the patient or family for one.
2. **Fusion beats come in two kinds, and the spike tells you which.** Pacing fusion has a pacing spike in front of a hybrid complex and means the device is sensing and working. Native fusion has no spike, occurs in the middle of a wide-complex tachycardia, and raises the probability of VT, though on its own it is not diagnostic. Complexes changing shape beat to beat with spikes in front of some is pacing fusion.
3. **A defibrillator does not treat every ventricular tachycardia.** It only acts on rhythms faster than a programmed cutoff, commonly well above 150. A slow VT at 124 can run untreated in a patient with a working device. Only interrogation shows what the device detected, and it was not done on this admission.
4. **Describe device activity in plain words at handoff.** What the complexes did, whether spikes were present, how long it lasted, what the rhythm was before and after, and whether the patient felt it. The pacing activity on this call does not appear anywhere in the hospital record, and a plain description is what gets a device checked.
5. **A good saturation says nothing about CO2.** He saturated 98 percent in the truck with a PCO2 that turned out to be 71. Capnography measures ventilation, which is the thing that was failing, and a heart failure patient with sleep apnea and COPD is exactly the patient to put it on.

Prepared from records obtained through the follow-up request process and de-identified for education. To request follow-up on a call you ran, fill out the request form at https://bcmca.github.io/followup/